

NADA MOHAMED

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Summary

Computer Science graduate (2026) specializing in Artificial Intelligence and Machine Learning, with a broad engineering background spanning intelligent agents, LLM fine-tuning, computer vision, full-stack development, blockchain, and high-performance computing. Have designed and shipped 15+ projects, from an award-winning facial-recognition system to a blockchain-based payroll microservice validated by 91 passing smart-contract tests. Comfortable across the stack in Python, Go, Java, and C/C++, and driven by turning research-grade ideas into practical, reliable software.

Education

Arab Academy for Science, Technology & Maritime Transport **Sep 2022 – 2026**

Bachelor of Computer Science (Dual Degree), Graduated with Honours – Cumulative GPA: 3.69 / 4.00 Cairo, Egypt

- Dual-degree program in collaboration with the University of Northampton (UK).
- Coursework: Machine Learning, Computer Vision, NLP, Algorithms & Optimization, Parallel Computing, Compilers, Databases, Robotics Applications, Software Engineering (SDLC).

Technical Skills

Programming Languages: Python, Java, C/C++, Go, JavaScript, TypeScript, Dart, PHP, R, Bash, SQL, HTML/CSS

AI / ML & Data: TensorFlow, PyTorch, Keras, Scikit-learn, NumPy, Pandas, OpenCV, Librosa, HuggingFace Transformers, Whisper; NLP, Generative AI & LLM fine-tuning (DeepSeek-Coder, FLAN-UL2), Computer Vision, Reinforcement Learning, Explainable AI (SHAP, LIME, Grad-CAM)

Web & Application: Flask, FastAPI, React, Tailwind CSS, shadcn/ui, Vite, JavaFX, Tkinter, Flutter, SwiftUI, PyWebView

Blockchain & Security: Hyperledger Fabric, Chaincode (Go), Privy MPC / threshold custody, X.509 PKI, Cryptography

Data, Cloud & DevOps: MySQL, AWS RDS, SQLite, Docker, Kubernetes, Git, GitHub Actions (CI/CD), launchd

CS Foundations: Algorithm design & optimization, Data structures, Parallel computing, Compilers (lexing/parsing), Robotics & embedded (Arduino), SDLC

APIs & Tools: Twilio, Groq, Pytest, Graphviz, L^AT_EX

Soft Skills: Team collaboration, Analytical thinking, Communication, Mentorship

Experience

SAIB Bank

Aug 2026 – Present

Fintech & Startup Partnerships Intern

Mohandeseen, Cairo, Egypt

- Working under the Head of Fintech & Startup Partnerships on how emerging technology can be applied to bank operations: streamlining products, raising productivity, and growing revenue.
- Authoring a business case for a chosen product, applying business-cycle analysis from rationale through rollout.
- Studying internal bank operations end to end: how the individual departments function and how they work together.

Dell Technologies

Aug 2025 – Sept 2025

Summer Intern

Cairo, Egypt

- Gained cross-department insight through sessions with Dell employees across sales, pre-sales, and engineering roles.
- Completed workshops on front-end development, containerization (Docker, Kubernetes), and software engineering practices.
- Applied design thinking to enhance Dell's onboarding experience; shadowed a project manager and software engineer to learn daily workflows.
- Led the Academy Project from concept to business proposal, strengthening technical and collaboration skills.

DEPI – Microsoft ML Engineering Curriculum

Nov 2024 – May 2025

AI/ML Engineer Intern

Zamalek, Egypt

- Completed a 7-month intensive AI/ML engineering program (Microsoft ML Engineering curriculum) covering NLP, generative AI, computer vision, and model deployment.
- Executed hands-on projects at each milestone and an end-to-end capstone (Jarvis AI Assistant) integrating multimodal workflows and intelligent agent orchestration.

Megz

Aug 2020 – Nov 2021

Computer Science Instructor

Cairo, Egypt

- Assisted 30+ students with programming fundamentals, improving proficiency in Python and Java.

- Supported the lead teacher in managing classrooms, grading, and tracking student progress; delivered personalized guidance that improved small-group performance by 20%.
- Mentored students in debugging and problem-solving, fostering a collaborative learning environment.

Projects

Payramid: Money Transaction Microservice

2025 – 2026

Go, Hyperledger Fabric, Privy MPC, Twilio, TypeScript

Graduation Project

- Built a blockchain-based payroll-orchestration and immutable-audit layer enabling employers to disburse salaries instantly with cryptographically verifiable proofs of payment, settling on Egypt's national rails (InstaPay, Meeza, ACH) within banking (Law 194/2020) and data-protection (Law 151/2020) regulation.
- Engineered the smart-contract layer as Hyperledger Fabric chaincode (Go) on a 3-organization RAFT network with MAJORITY endorsement; validated with an automated Go suite of 91 passing tests (100%) covering serialization integrity, double-spend prevention, and a full STRIDE threat model.
- Delivered offline reach via encrypted SMS relay (one-time withdrawal codes for feature phones) and non-custodial wallets using Privy threshold (MPC) key management; authored the complete IEEE 830 SRS and IEEE 829 test plan.

Artificial Intelligence & Machine Learning

Jarvis AI Assistant 🤖

Ongoing

Python, DeepSeek-Coder, FLAN-UL2, HuggingFace Transformers, Whisper

- Building a modular Jarvis-style AI assistant for code generation, image captioning, and voice interaction, with prompt-based instruction tuning and custom toolchain integration for autonomous task execution.
- Fine-tuned DeepSeek-Coder and FLAN-UL2 for multimodal encoder-decoder workflows; integrated NLP, Whisper-based speech recognition, and intelligent agent orchestration.

Oribu 🤖

Python, OpenCV, Siamese Neural Network, Twilio API

- Built an AI facial-recognition system using a Siamese neural network with contrastive loss for high-accuracy one-shot face verification, designed for criminal detection.
- Implemented real-time recognition with automated SMS alerts via the Twilio API. (*First Place AI/ML Project, AASTMT.*)

Music Genre Detection 🤖

Python, TensorFlow/Keras, Librosa, Flask

- Designed a CNN-LSTM-Attention deep-learning model classifying audio into 10 genres from mel-spectrograms, served through a Flask web interface.
- Added explainable-AI tooling (SHAP, LIME, Grad-CAM, Integrated Gradients), unit tests, and Sphinx-based API documentation.

CMAPSS Analytics Assistant 🤖

Python, FastAPI, Groq Llama 3.3 70B

- Built a NASA CMAPSS turbofan-degradation analytics service that parses FD001–FD004 telemetry, computes Remaining Useful Life (RUL), and ranks sensors by stability and informativeness.
- Exposed a FastAPI backend with an LLM-powered natural-language QA endpoint over failure-prediction signals.

Fera5 Invaders 🤖

Python, Pygame, NumPy, TensorFlow, PyTorch, Scikit-learn

- Developed an arcade-style space-shooter with a reinforcement-learning agent that autonomously learns to play the game.

Systems, Algorithms & Security

Parallel FFT Image Processing 🤖

C++, CMake, Parallel Computing

- Implemented 2D FFT-based image processing in C++ with sequential, partially-parallel, and fully-parallel variants to study parallel speedup, benchmarked through an automated CMake harness.

Interactive Knapsack Planner 🤖

C++, SFML 3.0, Dear ImGui

- Built an interactive algorithm visualizer for the 0/1 knapsack problem (greedy, brute-force, dynamic programming, branch-and-bound) with live DP-table animation and performance metrics.
- Added a Huffman-encoding module visualizing tree construction, bit-level encoding/decoding, and character-frequency analysis.

Simple Certificate Authority 🔒

Python, Cryptography, X.509 PKI

- Built a modular X.509 Certificate Authority supporting root-CA init, issuance, revocation (CRL), verification, renewal, and PKCS#12 export, with passphrase-encrypted keys and atomic database writes.

Text Editor (SCLPL) 🔒

Python, Tkinter, Pillow, Graphviz, Pytest

- Built a multi-tab editor with a full lexical and syntax-analysis pipeline for a custom toy language (SCLPL), including a hand-written lexer and parser for educational tooling.
- Added automatic language recognition, syntax highlighting, and parse-tree visualization.

OverClocked 🔒

Bash, Python, Zenity, Docker

- Built a cross-platform (Ubuntu/macOS) system-monitoring tool for hardware/software performance metrics with logging, visualization, and containerized deployment.

Applications, Automation & Games

Pixel Paradise 🔒

May 2024 – Feb 2025

JavaScript, PHP, HTML/CSS

- Lead contributor on a four-person team building a browser arcade platform, shipping four of the seven games end to end: Pong (with single-player and real-time two-player modes), Tetris, Snake, and Mario Bros.
- Built the site shell and front-end rendering, wiring game scores and session state into the shared leaderboard.

Job Apply Bot

Python, Groq Llama 3.3 70B, tectonic, launchd

- Built a compliant multi-ATS job-search automation that polls public APIs (Greenhouse, Lever, Ashby, Adzuna), filters new-grad SWE/ML roles by skill overlap and location, and logs matches to CSV.
- Auto-tailors a $\text{\LaTeX}$ resume per posting via an LLM and compiles per-job PDFs with tectonic; scheduled hourly via launchd.

Autonomous Pick-and-Place Robot 🔒

Arduino (C++), Flutter, Bluetooth

- Built the control system for an autonomous mobile robot (Arduino Uno) with differential drive, a 3-DOF servo arm/gripper, and ultrasonic + line-tracking sensors for a Pick-and-Place competition.
- Developed a Flutter Android app communicating over Bluetooth (HC-05) for manual and automatic navigation modes.

Trivia Game 🔒

Java, JavaFX, AWS RDS

- Built a trivia game with single-player and real-time multiplayer modes backed by an AWS-hosted database for room management, question delivery, and live score tracking.
- Implemented a multiplayer waiting room with real-time chat, “is typing” indicators, and audio/visual feedback cues using FXML interfaces.

Jarvis Desktop (Fluid Jarvis Whisper)

React, TypeScript, Tailwind, Flask, PyWebView

- Packaged the Jarvis assistant as an offline cross-platform desktop app with a React/Tailwind front end, Flask backend, and PyWebView wrapper built via Vite.

Certifications, Awards & Training

🏆 First Place, AI/ML Project Submission – AASTMT

🏆 Microsoft ML Engineer Course – DEPI (Digital Egypt Pioneers Initiative)

🏆 AI & Machine Learning Course – Universitat Autònoma de Barcelona

🏆 Computer Science Camp – Carnegie Mellon University

Languages

Arabic: Native

English: Fluent

Spanish: Intermediate